

POOJA AIREE

UG Second Year, CSE

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Academic Details

2020–2024 **B.E in Computer Science and Engineering.**
Chandigarh University (CU), Punjab
CGPA - 8.36

Present

2020 **All India Senior School Certificate Examination.**
City Convent School, Khatima, Uttarakhand (CBSE)
Percentage- 81.40% 99/100 marks in physical education subject

2018 **All India Secondary School Education Examination.**
SMSDutta Mem Nosegay Pub Sch, Khatima, Uttarakhand
(CBSE) Percentage- 74.16%

Internship

Summer 2021 **Proteomics Internship Orientation Program – IIT Bombay, Prof. Dr.Sanjeeva Srivastava**
Learning about mass spectrometry Data analysis and identification of proteins using different Search engine like- MASCOT, ProFound, Expasy, SEQUEST. Learned about Sample Preparation for Mass Spectrometry based Proteomics, and proteomic bioinformatics.

Present

Key Projects

- **Compiler for a C-like language** July, 2021
Developed a compiler for a C-like language in python, for MIPS instruction set architecture Supported major functionalities like function calls, **if-else statements**, **loops** and arithmetic expressions
- **A content-based music recommendation system** June, 2021
Followed [Oord et al, NeurIPS 2013] to implement a recommendation system based on audio signals, feedback of user likings and bag-of-word lyrics, using CNNs.
- **Assistive hand exoskeleton** May, 2021
Designed, modelled and fabricated Assistive hand exoskeleton to help a post-stroke patient with their rehabilitation process. Also participated Toycathon-2021 and selected in grand finale.
- **Newspaper Article Classifier** march, 2021
 - Developed models to classify newspaper articles into various categories.
 - Studied and implemented concepts of Machine Learning and Natural Language Processing like Word Tokenization, Bayes' Model and K-nearest neighbors.
- **An AI Bot Game** Jan, 2021
 - A two-player snake game, (human player & an AI bot), where one snake tries to run down the other one on a two-dimensional maze.
 - The bot had a two-step decision making process, existence of path, and traversal of path: used a greedy heuristic depth first search for validating existence of a path, followed by **greedy algorithm** to traverse the path.
- **Flappy Bird Game** September, 2020
A flappy bird game clone using Python and Pygame with user attracting graphics and sound.

RELEVANT COURSEWORK

Core Data Structures & Algorithms, Theory Of Computation*, Operating Systems*, Design and Analysis of Algorithms* (*-ongoing), Discrete Maths, Applied Game Theory, Logic in CS, Abstract Algebra, Probability & Statistics

Technical Proficiency

Operating Systems: Linux, Windows
Programming Languages: C/C++, Python, SQL, CSS, Java, JavaScript, PHP, HTML
Software/Devices: Arduino, Raspberry Pi, Unity Engine, Android Studio, AutoCAD, SolidWorks, Fusion 360, MATLAB, Bash, PyTorch, TensorFlow, SimScale

Achievements

2021, June: Selected among **Top 15 finalists** from all over the **world** in MEA Energy Innovation Hub's Global Virtual Hackathon, for doing Research in wind harvesting triboelectric nanogenerator.

2021, June: Participated in SmartIDEAthon-2021 Startup Competition and reached to round 3.

2021, March: Participated in team in the final of Start-a-Thon 2.0 Conducted by Sardar Patel Technology Business Incubator (SP-TBI) and secured a place in winners and received certificate and AWS Cloud Credits: **\$1,000 USD** (For Multiple Dot Refreshable Braille Display)

2021, Feb: Participated in team in National level Government competition Called Toycathon-2021 and **2** of our Project is selected for grand Finale.

2021, May: Secured 5th **Rank** in a science competition conducted during the Biology and Bioengineering Summer Camp 2021 By **IIT: Bombay**

2021, April: **Secured 1st** Position in Idea Bootcamp in competition with **71** Students, conducted by Chandigarh University.

2020, June: Awarded scholarship by Chandigarh University on qualifying the Common Entrance Test Scholarship Exam.

Publications

2019, December: **The Pendulum Wave (Published on Autodesk Instructables)**
The Pendulum Wave is a device that consists of a series of a pendulum with increasing periods that when simultaneously released, produce the effect of a changing transverse Wave that cycles back to the beginning conditions.
<https://www.instructables.com/The-Pendulum-Wave/>

EXTRA CURRICULAR ACTIVITIES

Launched an science YouTube Channel Called Science mode Laboratory to motivate students about science and **Crossed 1.9 million views.**
<https://www.youtube.com/channel/UCUgWP-UCYX4kz6JXRvH5cMQ>